

Applied Social Data Science

Coding Camp

Setup • Orientation

Elena Karagianni

karagiae@tcd.ie

 September 7–11, 2026

Department of Political Science, Trinity College Dublin

Schedule

10am - 1pm

- Monday: Introduction + R Basics I
- Tuesday: R Basics II + Good practices

10am - 12pm

- Thursday: Python basics
- Friday: How to write up and report on $\text{\LaTeX}$

Recap

In the last few days we looked at:

- Some basics of working in R
- Some basics of working in Python
- Some general good practices to follow when doing data science
 - Organizing your work
 - Being consistent in things like object naming

How to write up and report

- For your own projects
- For doing your class assignments

How to write up and report

- For your own projects
- For doing your class assignments

Today's class is intended to cover the basics and help orient your future learning.
We will look at:

How to write up and report

- For your own projects
- For doing your class assignments

Today's class is intended to cover the basics and help orient your future learning.
We will look at:

- R Markdown
- $\text{\LaTeX}$

- Combine code, results, and commentary in one document
- Export to many formats: PDF, Word, HTML
- Create reports, presentations, dashboards, and more

$\text{\LaTeX}$ is a free typesetting system. Unlike MS Word (WYSIWYG), it uses plain text and markup. It has a steep learning curve... so why use it?

- Free and open source

$\text{\LaTeX}$ is a free typesetting system. Unlike MS Word (WYSIWYG), it uses plain text and markup. It has a steep learning curve... so why use it?

- Free and open source
- Produces professional, consistent documents

LaTeX is a free typesetting system. Unlike MS Word (WYSIWYG), it uses plain text and markup. It has a steep learning curve... so why use it?

- Free and open source
- Produces professional, consistent documents
- Standard in many quantitative academic fields

$\text{\LaTeX}$ is a free typesetting system. Unlike MS Word (WYSIWYG), it uses plain text and markup. It has a steep learning curve... so why use it?

- Free and open source
- Produces professional, consistent documents
- Standard in many quantitative academic fields
- Powers outputs in many data science tools

Why you need $\text{\LaTeX}$

- R Markdown uses $\text{\LaTeX}$ “behind the scenes” to create PDFs. → You need a $\text{\LaTeX}$ installation to export PDFs from RStudio.

Why you need $\text{\LaTeX}$

- R Markdown uses $\text{\LaTeX}$ “behind the scenes” to create PDFs. → You need a $\text{\LaTeX}$ installation to export PDFs from RStudio.
- Because R Markdown files can be tricky to debug, in Stats I and II we’ll work directly in $\text{\LaTeX}$.

Why you need $\text{\LaTeX}$

- R Markdown uses $\text{\LaTeX}$ “behind the scenes” to create PDFs. → You need a $\text{\LaTeX}$ installation to export PDFs from RStudio.
- Because R Markdown files can be tricky to debug, in Stats I and II we’ll work directly in $\text{\LaTeX}$.
- Overleaf (<https://www.overleaf.com>) — an online, collaborative $\text{\LaTeX}$ editor

(yes, this is how these slides were created!)

Why you need $\text{\LaTeX}$

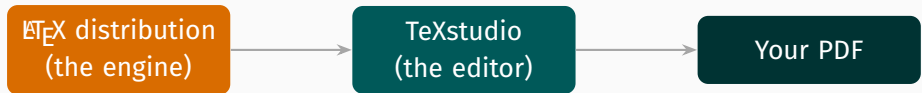
- R Markdown uses $\text{\LaTeX}$ “behind the scenes” to create PDFs. → You need a $\text{\LaTeX}$ installation to export PDFs from RStudio.
- Because R Markdown files can be tricky to debug, in Stats I and II we’ll work directly in $\text{\LaTeX}$.
- Overleaf (<https://www.overleaf.com>) — an online, collaborative $\text{\LaTeX}$ editor
- TexStudio — a free desktop IDE for $\text{\LaTeX}$ (<https://www.texstudio.org/>)

(yes, this is how these slides were created!)

Let's Get Set Up

- Two separate pieces of software, both free:
 - A **TeX distribution** — the “engine” that turns your document into a PDF
 - **TeXstudio** — the editor you'll actually type in
- It might take some time...the distribution is a large download
- Instructions for both **Windows** and **Mac** follow — use the one for your machine

Two Pieces, Working Together



- Same relationship as **R & RStudio**, or **Python & Jupyter**
- You write `.tex` files in TeXstudio; the distribution compiles them into a PDF behind the scenes
- **Windows** $\rightarrow$ **MiKTeX** • **Mac** $\rightarrow$ **MacTeX**

Before You Start: Checklist

-  Windows: ~1 GB to start (MiKTeX installs more *on demand*); Mac: ~5 GB (MacTeX installs everything upfront)
-  20-30 minutes, stable internet connection
-  Admin rights on your laptop
-  Close other applications before installing

Step 1: Install Your $\text{\LaTeX}$ Distribution

Windows — MiKTeX

1. Go to `miktex.org/download`
2. Download the **Basic Installer** (64-bit)
3. Run it → choose “Install for all users” (or “just me”)
4. When asked “Install missing packages on-the-fly”, choose **Yes** — important!
5. Click **Install** → **Finish**

Mac — MacTeX

1. Go to `tug.org/mactex`
2. Click **MacTeX Download** — saves a large **.pkg** (~5 GB)
3. Double-click the **.pkg**
4. Click **Continue** → **Agree** → **Install** (enter your Mac password)

Step 2: Install TeXstudio

Same steps on both operating systems:

1. Go to **texstudio.org** → **Download**
2. Windows: download the **.exe** installer → run it → **Next** → **Install** → **Finish**
3. Mac: download the **.dmg** → open it → drag **TeXstudio** into your **Applications** folder

 Install your $\text{\LaTeX}$ distribution **first** — TeXstudio needs it to compile anything.

Step 3: First Launch — Check the Connection

1. Open **TeXstudio**
2. Go to **Options** → **Configure TeXstudio** → **Commands**
3. Look at the **pdf $\LaTeX$** field — it should already show a path (auto-detected) with a green checkmark

- ✓ Green checkmark → TeXstudio found your distribution. You're ready to compile.
- ✗ Red mark or empty field → close and reopen TeXstudio (or restart your laptop) so it can detect the newly installed distribution.

Test It: Open the Template

1. Open **TeXstudio**
2. **File** → **Open** → select the template `.tex` file shared with you

Test It: Open the Template

1. Open **TeXstudio**
2. **File** → **Open** → select the template **.tex** file shared with you
3. Click the green **Build & View** arrow in the toolbar (or press **F5**)

Test It: Open the Template

1. Open **TeXstudio**
2. **File** → **Open** → select the template `.tex` file shared with you
3. Click the green **Build & View** arrow in the toolbar (or press **F5**)
4. Watch the **log panel** at the bottom: it should finish with no errors

Test It: Open the Template

1. Open **TeXstudio**
2. **File** → **Open** → select the template `.tex` file shared with you
3. Click the green **Build & View** arrow in the toolbar (or press **F5**)
4. Watch the **log panel** at the bottom: it should finish with no errors
5. A PDF appears in the preview pane on the right — that's your compiled document

Make a Change, Recompile

The loop you'll use for the rest of the course:

1. Edit some text in the template

Make a Change, Recompile

The loop you'll use for the rest of the course:

1. Edit some text in the template
2. Save the file (**Ctrl/Cmd + S**)

Make a Change, Recompile

The loop you'll use for the rest of the course:

1. Edit some text in the template
2. Save the file (**Ctrl/Cmd + S**)
3. Click **Build & View** again (or press **F5**)

Make a Change, Recompile

The loop you'll use for the rest of the course:

1. Edit some text in the template
2. Save the file (**Ctrl/Cmd + S**)
3. Click **Build & View** again (or press **F5**)
4. Check the preview pane — your change should now appear in the PDF

 Edit → Save → Build & View — repeat this loop every time you write $\text{\LaTeX}$.

You're Ready When...

- ✓ TeXstudio opens the template without errors
- ✓ Clicking **Build & View** produces a PDF
- ✓ Editing a line, saving, and recompiling updates the PDF

This concludes the Coding Camp. 🎉

Well done on making it through!

For additional queries: `karagiae@tcd.ie`